Deeper examination of some functions of the `datana` package

Examples

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This document expands on examples of different functions from the datana package (Salas-Eljatib and Campos 2025), implemented in R (R Core Team 2025). The main idea of such a report is to provide most of the utilities and versatility of the aforementioned package's functions.

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1 interp

1.1 About this function

The `interp` function dealing with interpolation. Cases on which interpolation is needed are usually those dealing with series data, such as: time series of economic, climate and biological varibles.

Nico: esta funcion solo sirve para datos de ahusamiento? o sirve para interpolar cualquier tipo de series?.

1.2 Example

In a forestry context, the `interp` function results specially handy for reviewing tree taper data. The idea is that with this function, some important variables (*e.g.,* commercial heights and/or diameters) can be imputed based upon the taper series data. Those variables are key for later on compute commercial volume.

Taper series data must fullfill the following:

1. Decreasing stem diameter along stem-height, *i.e.,* the largest stem-diameter is at the ground level (*i.e.,* stem-height equal to 0) and the smallest-diameter is at the tip of the tree (*i.e.,* stem-height equal to total height).

2. To always exists a section at stump height (generally at 0.3 m), dbh height, and between the mentioned values (generally 0.8m).

3. That the diameter outside bark is always larger than the diameter inside bark for the same height.

4. That the last section (the tip of the tree) has diameter equals 0 (zero) and $hl = h$.

1.2.1 Data

Loading tree taper data:

Nico: cargar mejor datos desde biometrics, los datos de eucalyptus regnans. Dejar primero a la columna de altura fustal, y luego diametro fustal.

```
df <- read.csv((./resources/ahusa.csv"))
head(df[, c("dl.cc", "hl")])
```

```
dl.cc   hl
1 11.02 0.05
2  8.46 1.30
3  8.10 1.58
4  7.86 2.37
5  6.90 3.16
6  6.70 3.95
```

This tree doesn't have a section at height 0.8 m, so this is interpolated.

```
library(datana)
## the height will be 0.8 m
```

```r
new.section <- 0.8
interpolated <- interp(x = df$dl.cc, y = df$hl, ylu = new.section)
interpolated$datares
```

```
x    y interpolated
1  11.020000 0.05        FALSE
2   9.378939 0.80         TRUE
3   8.460000 1.30        FALSE
4   8.100000 1.58        FALSE
5   7.860000 2.37        FALSE
6   6.900000 3.16        FALSE
7   6.700000 3.95        FALSE
8   6.000000 4.74        FALSE
9   5.500000 5.53        FALSE
10  4.400000 5.60        FALSE
11  3.500000 6.32        FALSE
12  2.820000 7.11        FALSE
13  0.000000 9.00        FALSE
```

Nico: agregar como se explica esta salida, que significa el x el t, etc, explicar que es "datares", etc.

2 plotrend

2.1 About this function

The function `plotrend` aims to provide a plot showing the main patter that might be occuring between two variables, expressed as the Y and X axis.

2.2 Example

Let's use the following dataframe

```
library(datana)
df <- datana::maple
head(df)
str(df)
```

```
tree  dbh leaf branch   bole  bark  total
1    1  9.1 0.67  21.30   9.94  1.51  33.42
2    2  9.1 0.54  20.30   6.21  0.63  27.68
3    3  9.9 0.72  19.26  13.93  1.69  35.60
4    4 10.2 0.60  21.77  17.68  1.60  41.65
5    5 16.7 2.90  28.47 101.94 11.37 144.68
6    6 17.2 2.39  26.94  90.26 12.35 131.94
'data.frame': 29 obs. of  7 variables:
 $ tree  : int  1 2 3 4 5 6 7 8 9 10 ...
 $ dbh   : num  9.1 9.1 9.9 10.2 16.7 17.2 17.6 25.1 25.4 25.9 ...
 $ leaf  : num  0.67 0.54 0.72 0.6 2.9 2.39 3.48 7.48 5.83 4.74 ...
 $ branch: num  21.3 20.3 19.3 21.8 28.5 ...
 $ bole  : num  9.94 6.21 13.93 17.68 101.94 ...
 $ bark  : num  1.51 0.63 1.69 1.6 11.37 ...
 $ total : num  33.4 27.7 35.6 41.6 144.7 ...
```

We will fit the following simple regression model

```
m1 <- lm(leaf ~ dbh, data = df)
summary(m1)
```

```
Call:
lm(formula = leaf ~ dbh, data = df)

Residuals:
     Min       1Q   Median       3Q      Max 
-10.0875  -1.7636  -0.3398   1.7989  10.7639 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -6.14981    2.39450  -2.568   0.0161 *  
dbh          0.53746    0.07484   7.181 1.01e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Residual standard error: 4.537 on 27 degrees of freedom
Multiple R-squared:  0.6564, Adjusted R-squared:  0.6436
F-statistic: 51.57 on 1 and 27 DF,  p-value: 1.008e-07
```

A customary question in regression analysis is to detect any departure from the assumption of the equal variance of the residual. In such a case, the function `plotrend` is suitable, to assess the pattern of the residuals agains any predictor variable (Fig. 1) and against the fitted values for the response variable (Fig. 2), too.

Nico: puedes agregar el label de cada figura en el texto, de forma automatica. Ademas, hacer que el siguiente codigo no se imprima, gracias

```r
df$yaju<-fitted(m1)
df$eaju<-residuals(m1)
head(df)
tail(df)
```

```r
datana::plotrend(x = df$dbh, y = residuals(m1), ylab="Residuals (Kg)", xlab = "Diameter at
↪  breast-height (cm)")
abline(h = 0)
```

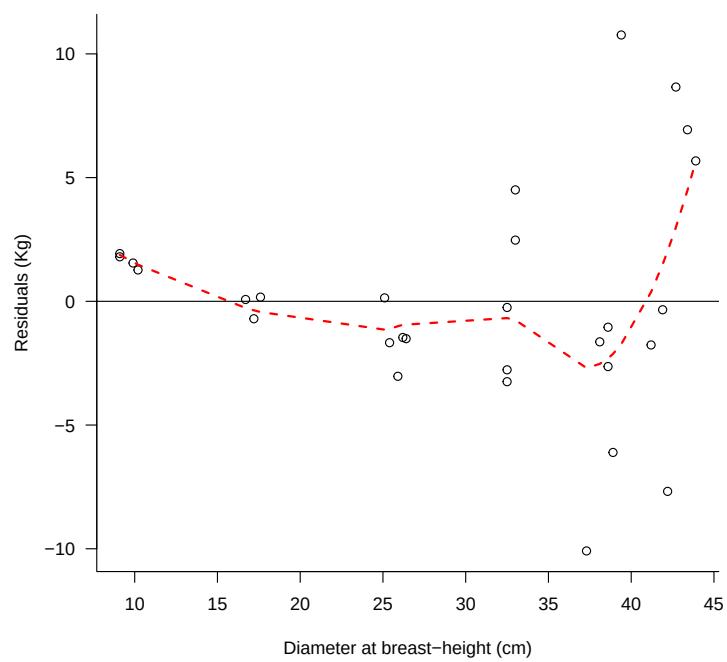

Figure 1: Scatterplot between the residuals of the fitted model and the predictor variable of the model, using the `plotrend` function.

```r
datana::plotrend(x = fitted(m1), y = residuals(m1), xlab = "Fitted values (Kg)",
↪  ylab="Residuals (Kg)")
abline(h = 0)
```

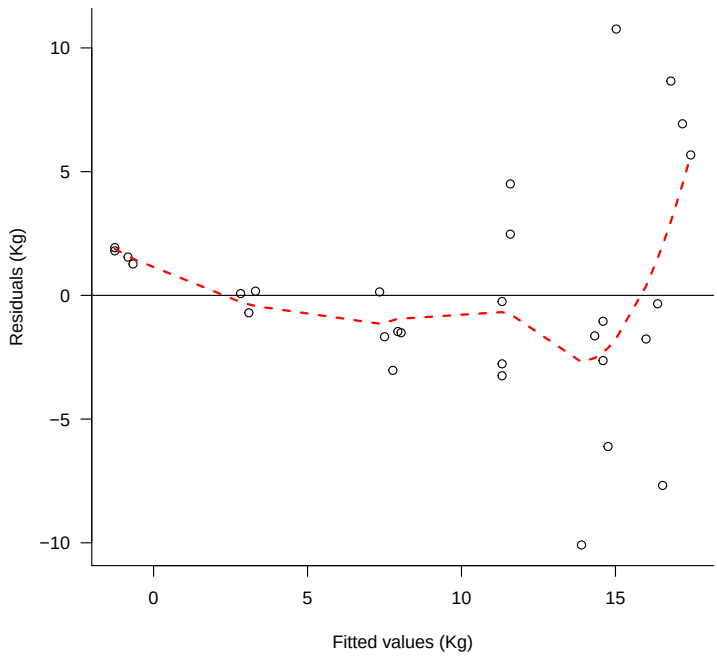

Figure 2: Scatterplot between the residuals of the fitted model and the fitted values for the predictor variable of the model.

3 descstat

3.1 about this function

3.2 Datos para el ejemplo

Se cargan los datos de ahusamiento de un árbol

```
library(datana)
df <- datana::maple
head(df)
descstat(df)
```

```
tree  dbh leaf branch   bole  bark  total
1    1  9.1 0.67  21.30   9.94  1.51  33.42
2    2  9.1 0.54  20.30   6.21  0.63  27.68
3    3  9.9 0.72  19.26  13.93  1.69  35.60
4    4 10.2 0.60  21.77  17.68  1.60  41.65
5    5 16.7 2.90  28.47 101.94 11.37 144.68
6    6 17.2 2.39  26.94  90.26 12.35 131.94

            tree    dbh   leaf  branch     bole    bark    total
n         29.000 29.000 29.000  29.000   29.000  29.000   29.000
Minimum    1.000  9.100  0.540  19.260    6.210   0.630   27.680
Maximum   29.000 43.900 25.790 482.820 1154.170 118.690 1781.140
Mean      15.000 29.948  9.946 139.828  508.944  52.972  712.001
Median    15.000 32.500  8.550 121.290  588.350  57.220  759.600
Std. Dev.  8.515 11.456  7.600 113.904  336.156  35.199  479.196
CV(%)     56.765 38.252 76.409  81.460   66.050  66.449   67.303
```

Se ajusta un modelo lineal

```
m1 <- lm(leaf ~ dbh, data = df)
m1
```

```
Call:
lm(formula = leaf ~ dbh, data = df)

Coefficients:
(Intercept)          dbh
    -6.1498       0.5375
```

Usando la función

```
datana::plotrend(x = df$dbh, y = residuals(m1), xlab = "DBH (cm)")
abline(h = 0)
```

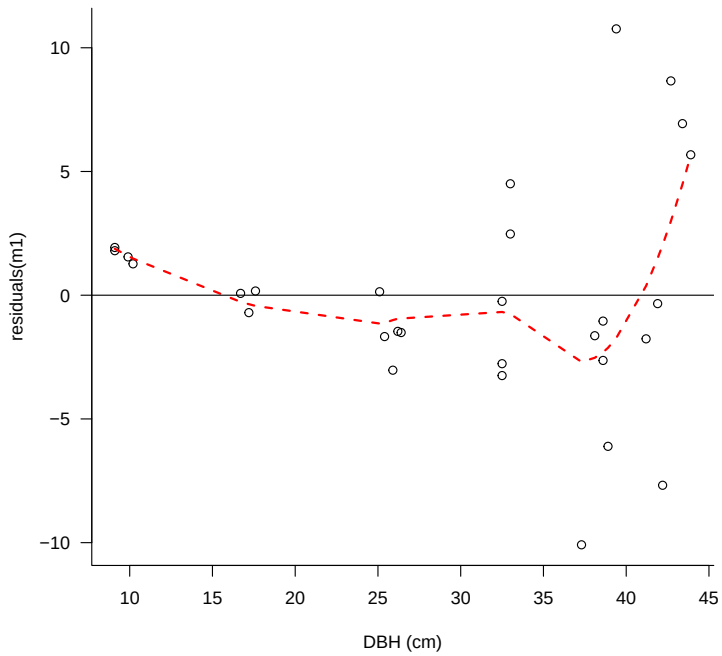

Figure 3: Example plot generated with the `plotrend` function.

References

R Core Team (2025). *R: A language and environment for statistical computing*. `http://www.R-project.org`. R Foundation for Statistical Computing. Vienna, Austria.

Salas-Eljatib, C. and N. Campos (2025). `datana`*: Data and functions to accompany Análisis de datos con R*. R package version 1.1.4. DOI: `10.32614/CRAN.package.datana`. URL: `https://cran.r-project.org/package=datana`.